Glossary

Institutional Memory — DAQ Condition Intelligence, one plain sentence per term, no formulas

RMS (root mean square)	A single number that summarises how strong a signal is overall, averaged over a window, the way "average loudness" summarises a sound clip.
Peak	The single largest value (in either direction) seen anywhere in the window — the biggest spike.
Crest factor	How spiky a signal is: the peak compared to the overall strength (RMS). A high crest factor means occasional sharp spikes on top of a calmer background; a low one means the signal is more evenly rough throughout.
Kurtosis	A measure of how "spiky and rare" the extreme moments in a signal are, compared to a smooth, evenly varying signal. Higher kurtosis often shows up around impacts or sudden events.
Skewness	A measure of whether a signal leans more toward high values or low values overall, rather than being evenly balanced around its average.
Spectral band energy (low / mid / high)	How much of a signal's energy falls into a low, medium, or high range of vibration frequency — similar to how a bass/mid/treble display shows where a sound's energy sits, rather than just how loud it is overall.
Spectral centroid	Roughly, the "center of balance" of a signal's frequency content — whether the energy is concentrated toward lower or higher frequencies on average.
Sampling rate	How many readings per second a sensor recorded. This is measured from each recording's own time column, never assumed, because getting it wrong would make every frequency-based measurement meaningless.
Window	A short, fixed-length stretch of consecutive sensor readings (2,048 samples here) that all the per-window measurements are computed from.
Baseline	The "normal" profile for one specific condition and one specific channel, built only from recordings that passed the quality check for that exact condition.
Mahalanobis distance	A way of measuring how unusual a window's set of measurements is compared to its condition's baseline, that accounts for the fact some measurements naturally vary more than others.

Anomaly score	The number that comes out of the Mahalanobis distance calculation for a window. A higher score means the window looks less like the normal pattern for its condition.
Finding	A window that scored high enough on anomaly to be worth mentioning, together with the plain-language reason and the exact file/channel/range it came from.
Condition	One specific combination of spindle speed, feed rate and depth of cut (S, F, D) that the machine was run under during a set of recordings.
Canonical schema	The expected seven-column layout every recording should have. A recording that matches it is called "canonical" and is used in analysis.
Quarantine	Setting a file aside, with a logged reason, instead of using it or trying to repair it, when it doesn't match the canonical schema.
Citation ([Finding N])	A marker in an answer that points to the exact Finding it's based on, so every specific claim can be traced back to real data.
Raw voltage (V)	The actual unit every channel is recorded in — a direct sensor output that hasn't been converted into a physical unit like g-force, newtons or amps. Confirmed from the “Unit” row present in every recording.
Latin-square design	A way of planning an experiment so every combination of two settings (here, speed and feed) is paired with a different value of a third setting (depth of cut) exactly once — used across all 25 conditions in this dataset.

Note for integration: this maps directly onto the existing glossary field in `config/domain.yaml`, which `llm/prompt_builder.py` already scans a question against and auto-injects when a term appears — reuse that mechanism as-is:

```
glossary:
  - term: RMS
    definition: "A single number that summarises how strong a signal is overall, averaged over a window."
  - term: Mahalanobis distance
    definition: "A way of measuring how unusual a window looks compared to its condition's normal pattern."
  # ...one entry per term above
```

Source: signal/features.py (feature list), analytics/anomaly.py (mahalanobis-diagonal-v3), and README.md §2–§3.